# Library Management System

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MAGAZINE

"A well-managed library is the gateway to knowledge – and an efficient library management system is the key that unlocks it."



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# Introduction to Library Management Systems


In today's quickly changing digital world, libraries have changed from just places that store books into lively centers of knowledge and learning. At the center of this change is the Library Management System (LMS) – a software program designed to make the hard jobs of organizing books, lending them out, keeping track of stock, and helping users easier. An LMS not only automates everyday tasks but also improves the user experience by giving easy access to both digital and physical materials. From keeping track of borrowed books and managing memberships to supporting digital collections and online catalogs, these systems help librarians handle resources better and help users find information faster. By combining technology with traditional library work, LMSs play an important role in making sure libraries stay useful, easy to access, and friendly in today's information age.

# Book Class functions explained

## 1 BOOK

It is a parameterized constructor with default arguments used to initialize the values of the variables (book name, category, author, and borrowed status) in case the user didn't enter them and they can also be manually initialized

## 2 VALIDITY

It takes a string (category) as a parameter and checks whether the entered category is available in the array of categories in the function (fiction, romance, children, non-fiction, horror, history) by iterating over the array. If the category is available, the function returns true; otherwise, it returns false. It iterates 12 times because each category is written twice: one time in uppercase and the other in lowercase to handle all exceptions and not return false even if the enetered category is correct

## 3 SETTERS (TITLE, AUTHOR)

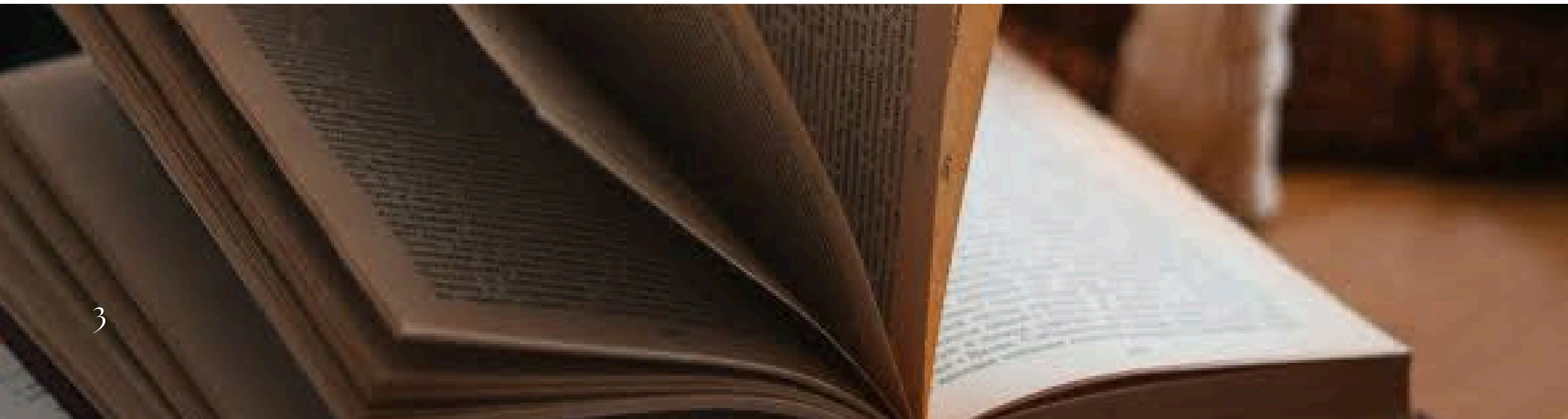
Using setters along with the parameterized since the constructor can only be called once, so the setters are used just in case the user wanted to change any of the variables, it would be possible. It is used for later modification.

## 4 SETCATEGORY

it takes a string (c for category) as a parameter. Then, calls the function "Validity" to check whether the entered category is available before setting the value of category. If the "Validity" function returned true, the value of category is set to c; otherwise, an error message will be printed to the user as invalid and the category will be automatically set to unspecified.

## 5 GETTERS (TITLE, AUTHOR, CATEGORY)

Getters are used to get the values of the variables. "const" is used to guarantee that there won't be any changes made in the objects thy will just show the values of the variables. Yet, if there was a change made in the setters, the getters will be updated accordingly; it doesn't block the changes made elsewhere, it just guarantees that the getter itself won't make changes.



# Library Class functions explained

## 1 LIBRARY

This is the constructor.
It sets the starting number of books (count) to 0. So, when the library is first created, it's empty.

## 2 ADDBOOK

Adds a book tob the library. It puts the new book at the next free sopt in the array, increasing the count by 1. If the library has already a 100 books, it displays a message that the library is already full.

## 3 REMOVEBOOK

It removes a book based on its title. It searches the array for a book with a matching title. Once found, it shifts all the books after that one step to the left; then, decreases the count by 1. If no book was found with that title, it displays “Book not found”.

## 4 SEARCHBYTITLE

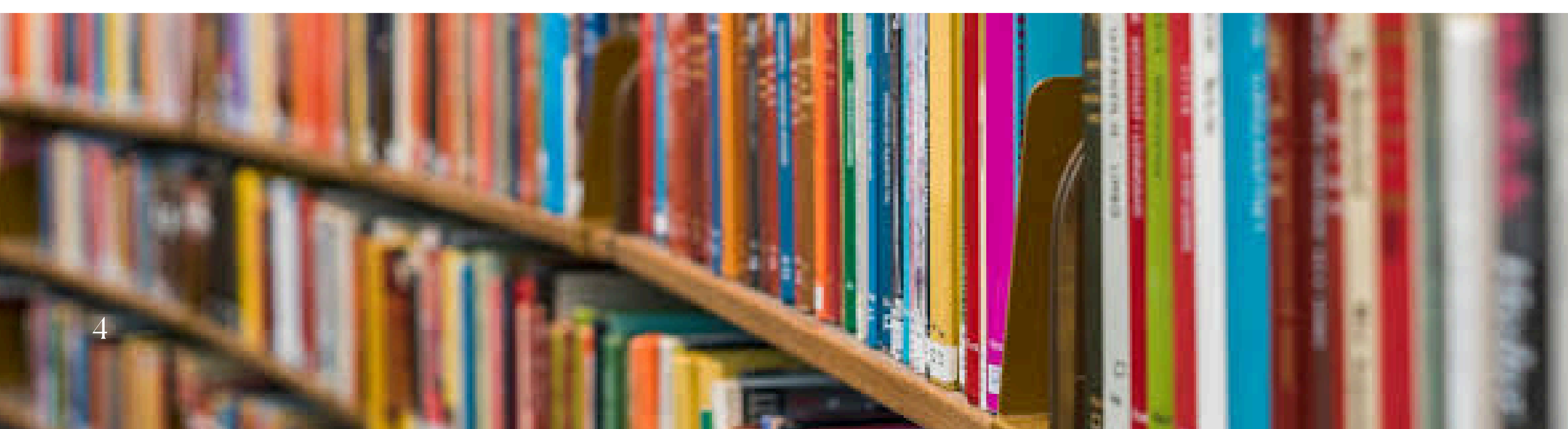
It searches the library for any books with a matching title and returns a formatted string with the book’s details once found. If multiple books have the same title, it returns all of them. If none are found, it returns an empty string.

## 5 GETAVAILABLEBOOKS

It goes through all books in the array and checks if each book is available (status=false). If yes, it adds the book’s information to a result string and returns the list of available books.

## 6 GETBORROWEDBOOKS

It has a similar mechanism to getAvailableBooks(), but it checks if the book is borrowed (status=true) and returns the list of borrowed books.



# User Class functions explained

## 1 USER

A default constructor that sets the ID number to zero and sets username and password to an empty string.

## 2 USER(INT ID,STRING UNAME, STRING PWD)

parametrized constructor that uses the data entered by the user and stores it.

## 3 GETUSERID

a function used to access the user's id as it is protected data and cannot be accessed from outside the class.

## 4 GETUSERNAME

function to return username as it is also protected data.

## 5 AUTHENTICATE

function used to check if the provided username and password match the stored ones so the user can gain access to the system.

# Librarian Class functions explained

## 1 LIBRARIAN

This is the constructor that initializes a Librarian object with an ID, username, and password. It calls the base User constructor, assigns the customer array pointer, references the shared customer count, and sets the maximum number of customers

## 2 ADDBOOKTOLIBRARY

adds a new book t the library using lib.addBook(b) calls Library class's method directly.

## 3 REMOVEBOOKFROMLIBRARY

It removes a book based on its title using Library class function removeBook

## 4 VIEWALLBOOKS

it displays all books (available and borrowed) using lib.getAvailableBooks() and lib.getBorrowedBooks().

## 5 SEARCHBOOKINLIBRARY

Search for books using partial or full title using Library::searchByTitle and prints result.

## 6 ADDCUSTOMER

Registers a new customer and checks (using an if condition) if maxCustomers is not exceeded, then adds to array and increments count.

## 7 REMOVECUSTOMER

Removes a customer by username.

## 8 VIEWCUSTOMERBORROWEDBOOKS

Lists all customers and their borrowed books by looping over all customers and prints ID, username, and borrowed book

# Customer Class functions explained

## 1 CUSTOMER()

A default constructor that calls the user class default constructor, and it also sets the borrowed count to zero.

## 2 CUSTOMER(INT ID, STRING UNAME, STRING PWD);

Parameterized constructor that uses the user class constructor to initialize all data needed in this class

## 3 BORROWBOOKFROMLIBRARY

This function checks first the number of books the user borrowed, and if less than 5, the book the customer wants is added to an array and the number of books borrowed is incremented. And if the user already borrowed 5 books, the function returns false which prevents the user from borrowing the 6th book.

## 4 RETURNBOOKTOLIBRARY

When the user wants to return a book, this function loops through the books the customer borrowed and compares their titles to the one the user entered as the book to be returned. If the title was found, the book title will be removed from the array and the number of books borrowed will be decremented. If title not found, the user receives a message that this book was not on the list to begin with.

## 5 LISTBORROWEDBOOKS

This function loops through the array of borrowed books and displays them on the screen.

# Extra Functions

## 1 SETUPLIBRARIAN

this function creates and returns a Librarian object using its constructor with a fixed ID, username, and password and gives them access to the list of customers and the system's customer limits.

## 2 LIBRARIAN VIEW

It's a menu-based function that allows the librarian to view the options of what they can do using a do-while loop that'll keep repeating until the librarian chooses to logout, which means that as long as the choice is a valid one and the condition of the loop is true, it'll keep going. It has 8 options: 1. add books, 2. remove book, 3. view all books, 4. search book, 5. add customer, 6. remove customer, 7. view all customers, 8. logout. It uses switch cases based on the number of the choice picked by the librarian and based on that number, the function calls the related class function to perform the task

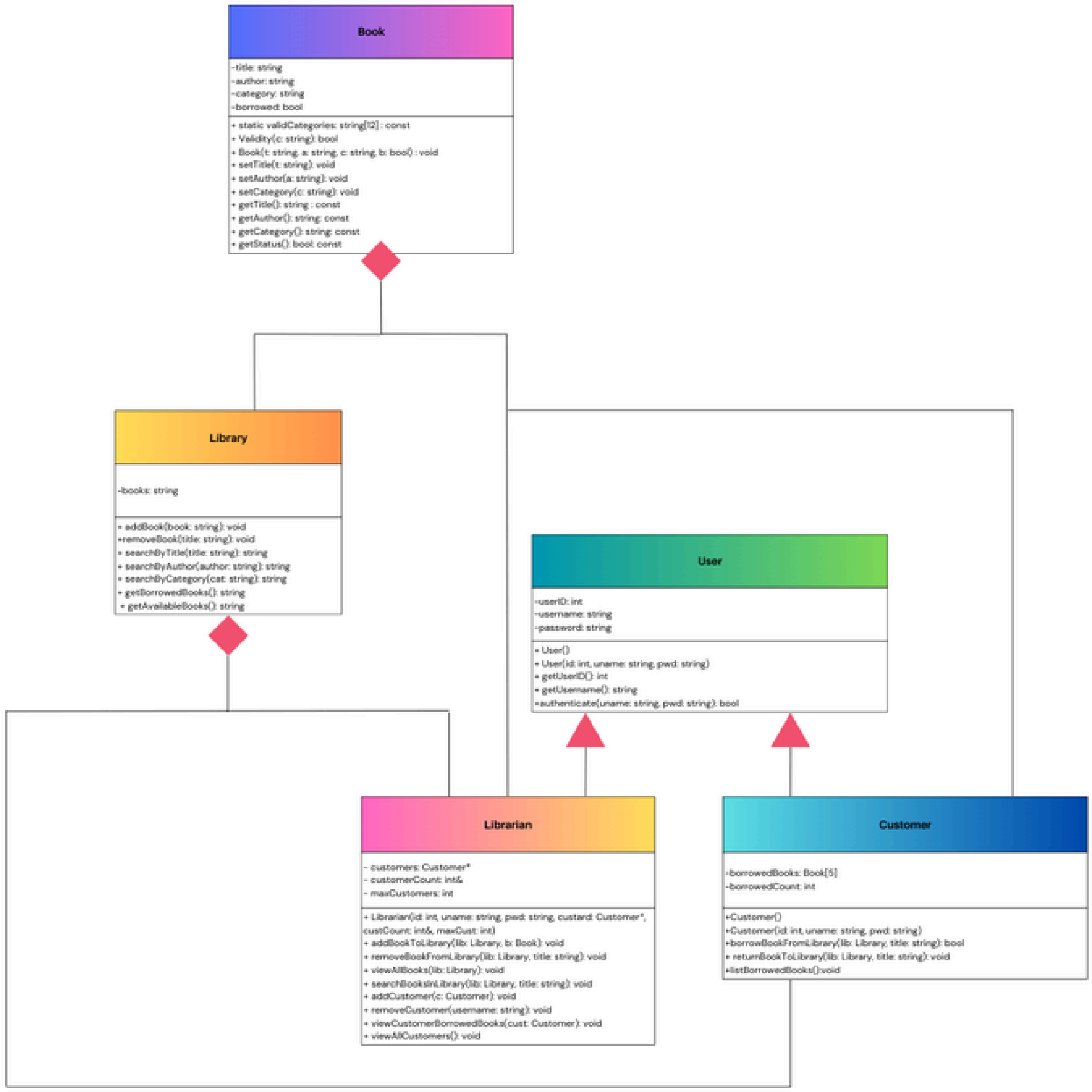
## 3 LOGINCUSTOMER

This function logs in a customer by checking if the entered username and password match any of the existing customers in the system by using a for-loop that goes through the list of customers and calls authenticate class function to check if the username and password match. If a match is found, the function returns a pointer to that customer. This means login was successful. If not, there will be displayed an error message and returns a nullptr to show that the login process failed.

## 4 CUSTOMERVIEW

It is also a menu-based function but for customers. It uses a do-while loop that'll keep repeating as long as the customer keeps choosing valid option until they choose to logout. It has 6 options: 1. view all available books, 2. search book, 3. borrow book, 4. return book, 5. my borrowed books, 6. logout. A switch case has been used to divert between cases based on the input number used for each case and based on that number, the function calls the related class function to perform the task

# Updated UML




## WORKLOAD DISTRIBUTION

Haya Moussa (24-101252): Book class
Zyad Walid (24-10301): Librarian class
Omar Khattab (24-101227): User and Customer classes
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